
5: Keeping the Working Environment

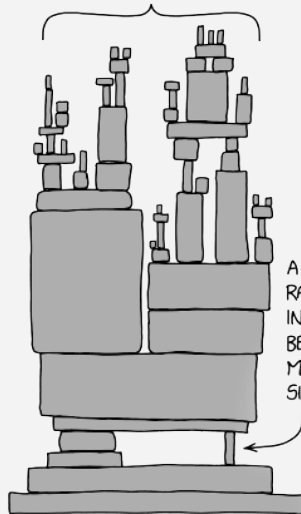
Capturing the working environment with conda and apptainer

Kosmas Hench



2025-05-28

ALL MODERN DIGITAL
INFRASTRUCTURE



A PROJECT SOME
RANDOM PERSON
IN NEBRASKA HAS
BEEN THANKLESSLY
MAINTAINING
SINCE 2003

SUPPORTING INFORMATION TEXT

R PACKAGES USED. The following R packages (including their dependencies) were used directly: ape (5.4.1) (1), BAMMtools (2.1.7) (2), bookdown (0.19) (3), cowplot (1.1.1) (4), FastEPRR (1.0) (5), furrr (0.1.0) (6), geiger (2.0.7) (7), GenomicOriginsScripts (0.0.1.1) (8), geomfactory (9), ggforce (0.3.2.9000) (10), ggfortify (0.0.0.9000) (11), ggiraph (0.7.0) (12), ggplotify (0.0.5) (13), ggpointdensity (0.1.0) (14), ggraph (2.0.5.9000) (15), ggraphify (0.0.0.9000) (16), ggstance (0.3.4) (17), ggtext (0.1.1) (18), ggtree (2.5.1.9002) (19), glue (1.4.2) (20), grid (4.0.3) (21), gtable (0.3.0.9000) (22), gtools (3.8.2) (23), hypoimg (1.0.0.1) (24), igraph (1.2.6) (25), IRanges (2.22.2) (26), kableExtra (1.3.0.9000) (27), kableExtra (1.3.0.9000) (28), kableExtra (1.3.0.9000) (29), kableExtra (1.3.0.9000) (30), kableExtra (1.3.0.9000) (31), kableExtra (1.3.0.9000) (32), kableExtra (1.3.0.9000) (33), kableExtra (1.3.0.9000) (34), kableExtra (1.3.0.9000) (35), kableExtra (1.3.0.9000) (36), kableExtra (1.3.0.9000) (37), kableExtra (1.3.0.9000) (38), kableExtra (1.3.0.9000) (39), kableExtra (1.3.0.9000) (40), kableExtra (1.3.0.9000) (41), kableExtra (1.3.0.9000) (42). The precise R configuration including package versions as well as dependencies is available in the supplementary git repository using the renv package.

good luck trying to re-assemble
that working environment...

SOFTWARE VERSIONS. The following software versions were used: admixture (1.3.0) (43), ASTRAL-III (5.7.5) (44), BCFtools (1.9) (45), BLAST (2.6.0) (46), Braker (1.16.0) (47), Dsuite (0.4.138) (48), gatk (4.0.8.1) (49), GBlocks (0.9.1b) (50), gemma (0.98) (51), GEVA (v1.0.0) (52), IQ-TREE (2.1.2) (53), MAFFT (v7.475) (54), msmc2 (2.0.0) (55), nextflow (0.31.1.4886) (56), PGDSpider (2.1.1.5) (57), plink (v1.90b4) (58), Python (2.7.15) (59), R (3.5.2, for analysis) (60), R (4.0.2, for visualization), RAXML-NG (v1.0.2) (61), samtools (1.9) (62), shapeit (v2.1837) (63), truffle (v1.38) (64) and vcftools (0.1.14 & 0.1.15) (65).

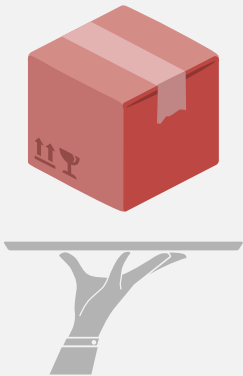
SUPPORTING INFORMATION TEXT

R PACKAGES USED. The following R packages (including their dependencies) were used directly: ape (5.4.1) (1), BAMMtools (2.1.7) (2), bookdown (0.19) (3), cowplot (1.1.1) (4), FastEPRR (1.0) (5), furrr (0.1.0) (6), geiger (2.0.7) (7), GenomicOriginsScripts (0.0.1.1) (8), geomfactory (9), ggforce (0.3.2.9000) (10), ggfortify (0.0.0.9000) (11), ggiraph (0.7.0) (12), ggplotify (0.0.5) (13), ggpointdensity (0.1.0) (14), ggraph (2.0.5.9000) (15), ggraphify (0.0.0.9000) (16), ggstance (0.3.4) (17), ggtext (0.1.1) (18), ggtree (2.5.1.9002) (19), glue (1.4.2) (20), grid (4.0.3) (21), gtable (0.3.0.9000) (22), gtools (3.8.2) (23), hypoimg (1.0.1.1) (24), igraph (1.2.6) (25), IRanges (2.22.2) (26), kableExtra (1.3.0.9000) (27), kableExtra (1.3.0.9000) (28), kableExtra (1.3.0.9000) (29), kableExtra (1.3.0.9000) (30), kableExtra (1.3.0.9000) (31), kableExtra (1.3.0.9000) (32), kableExtra (1.3.0.9000) (33), kableExtra (1.3.0.9000) (34), kableExtra (1.3.0.9000) (35), kableExtra (1.3.0.9000) (36), kableExtra (1.3.0.9000) (37), kableExtra (1.3.0.9000) (38), kableExtra (1.3.0.9000) (39), kableExtra (1.3.0.9000) (40), kableExtra (1.3.0.9000) (41), kableExtra (1.3.0.9000) (42). The precise R configuration including package versions as well as dependencies is available in the supplementary git repository using the renv package.

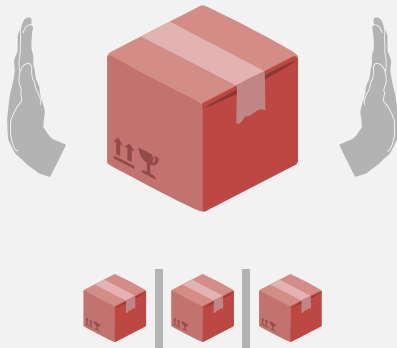
good luck trying to re-assemble
that working environment... 

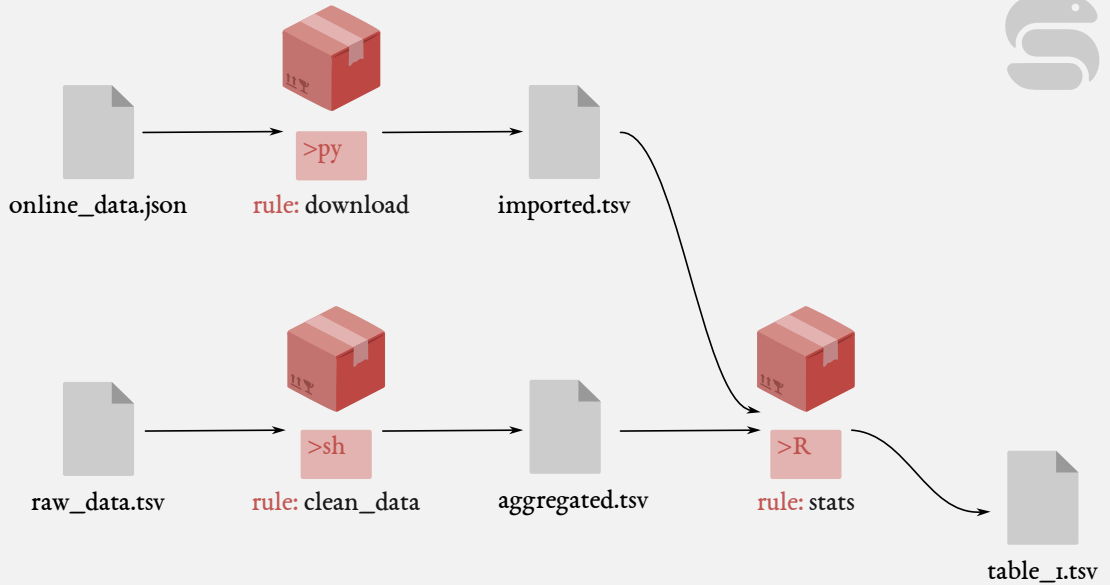
SOFTWARE VERSIONS. The following software versions were used: admixture (1.3.0) (43), ASTRAL-III (5.7.5) (44), BCFtools (1.9) (45), BLAST (2.6.0) (46), BWA (0.7.17-r1188) (47), Dsuite (0.4 r38) (48), gatk (4.0.8.1) (49), GBlocks (0.9.1b) (50), gemma (0.98) (51), GEVA (v1.0.0) (52), IQ-TREE (2.1.2) (53), MAFFT (v7.475) (54), msmc2 (2.0.0) (55), nextflow (0.31.1.4886) (56), PGDSpider (2.1.1.5) (57), plink (v1.90b4) (58), Python (2.7.15) (59), R (3.5.2, for analysis) (60), R (4.0.2, for visualization), RAXML-NG (v1.0.2) (61), samtools (1.9) (62), shapeit (v2.r837) (63), truffle (v1.38) (64) and vcftools (0.1.14 & 0.1.15) (65).

portability



isolation

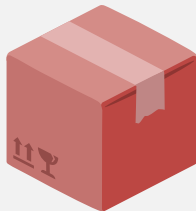




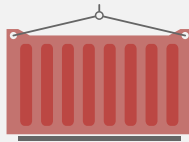
environment



instructions



container



real deal



A top-down view of a kitchen counter with various ingredients and a person cracking an egg. In the top left, a gold-colored plate holds several red raspberries. In the top center, another gold-colored plate holds several blackberries. To the right, a brown cloth holds several brown eggs. In the bottom center, a person's hands are cracking an egg into a bowl of white batter. A gold-colored spoon is in the bowl. To the right of the bowl, two eggshells are visible. In the bottom right corner, a wooden-handled knife and a small bowl of strawberries are partially visible. The counter is light-colored with some crumbs.

CONDA

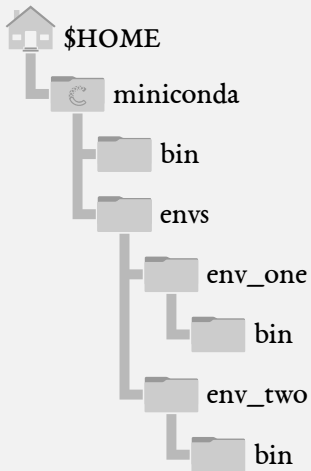
Image by Michelle Henderson



“Conda is an open-source, cross-platform, language-agnostic package manager and *environment management system*.”

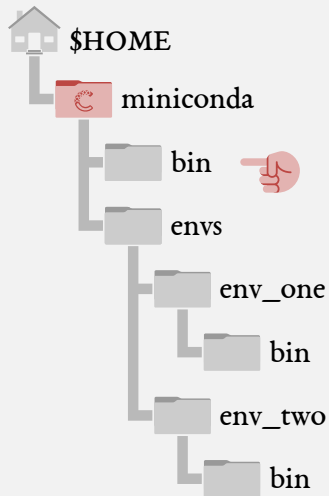


“Conda is an open-source, cross-platform, language-agnostic package manager and *environment management system*.”



\$ |

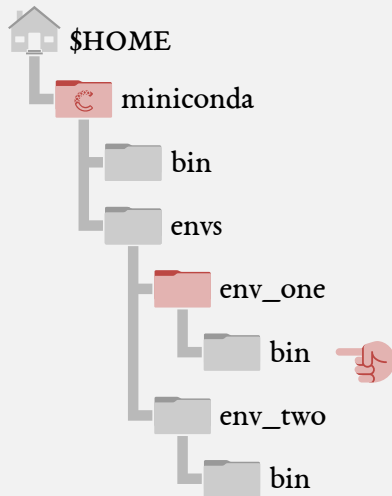
PATH=\$PATH



```
$ conda activate
```

```
(base) $ |
```

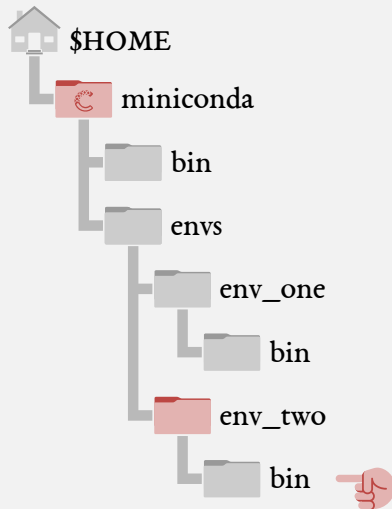
```
PATH=$HOME/miniconda/bin:$PATH
```



```
$ conda activate env_one
```

```
(env_one) $ |
```

```
PATH=$HOME/miniconda/envs/env_one/bin:$PATH
```



```
$ conda activate env_two
```

```
(env_two) $ |
```

```
PATH=$HOME/miniconda/envs/env_two/bin:$PATH
```

```
$ R --version
```

```
R version 4.4.3 (2025-02-28) – "Trophy Case"
```



```
$ conda run -n env_one R --version
```

```
R version 4.2.3 (2023-03-15) – "Shortstop Beagle"
```



```
$ conda run -n env_two R --version
```

```
R version 4.3.0 (2023-04-21) – "Already Tomorrow"
```





```
# conda env create -f r_base.yml
name: r_base
channels:
  - conda-forge
  - bioconda
  - defaults
dependencies:
  - conda-forge::r-base=4.4.2
```

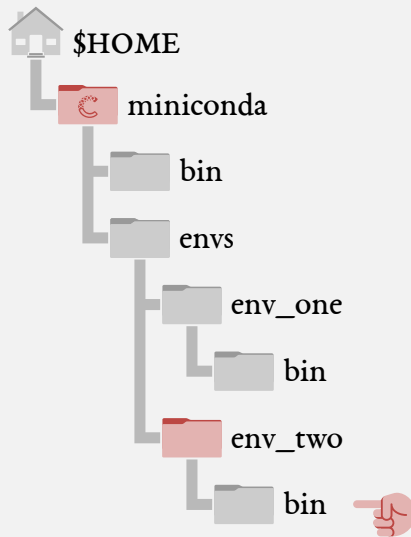
```
$ conda env create -f <file_name>.yaml
```

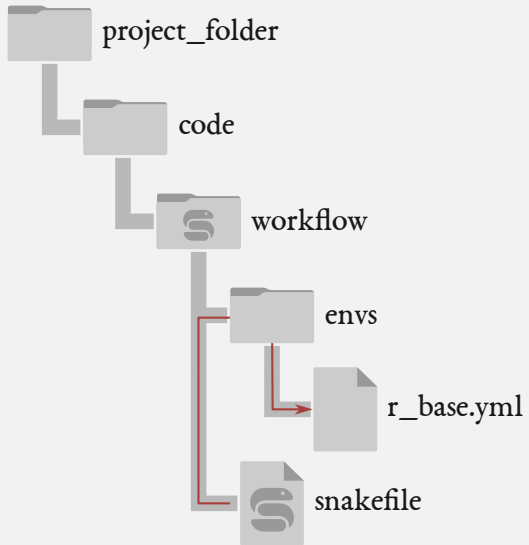


```
$ snakemake -c 1 --use-conda
```




```
rule some_job:  
    conda: "env_two"  
    shell:  
        ""  
  
    <some code>  
        ""
```





```
rule rjob:
```

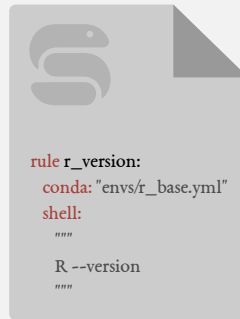
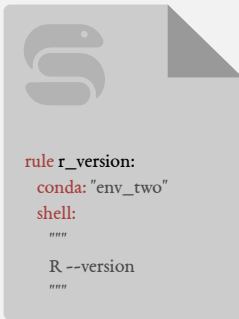
```
conda: "envs/r_base.yml"
```

```
shell:
```

```
"""
```

```
Rscript R/script.R
```

```
"""
```

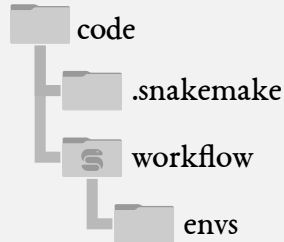


<local default>



\$HOME

miniconda



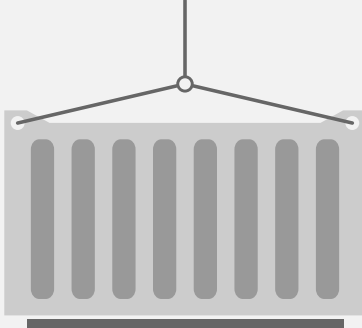


APPTAINER



Image by Caitlyn-de Wild

same, same



but different



docker

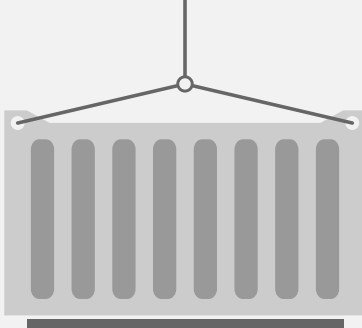


apptainer



podman

same, same



but different



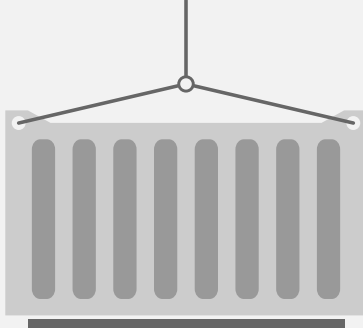
docker



apptainer



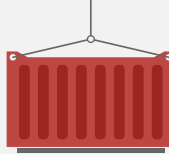
podman



apptainer



container.sif



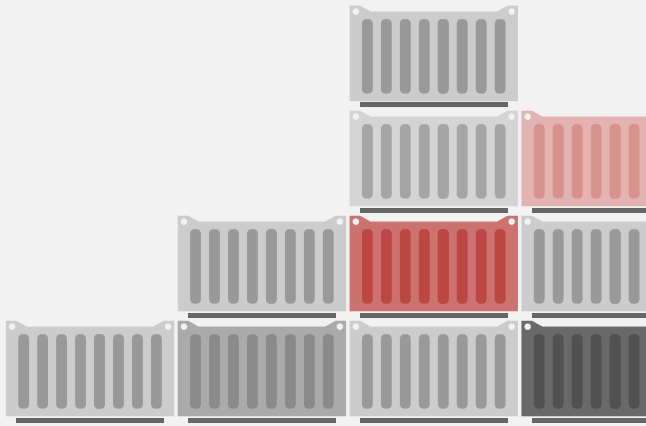
dockerhub.com



quay.io



[rocker \(R\)](#)




```
$ apptainer pull <container_url>:<container_version>
```

```
$ apptainer pull docker://rocker/tidyverse:4
```

— download —



tidyverse_4.sif

```
$ apptainer pull <container_url>:<container_version>
```

```
$ apptainer pull docker://rocker/tidyverse:4
```

— download —



tidyverse_4.sif



```
$ apptainer shell <apptainer_args> <container_url>:<container_version>
```

```
Apptainer> |
```



— enter —



```
$ apptainer pull <container_url>:<container_version>
```

```
$ apptainer pull docker://rocker/tidyverse:4
```

— download —



tidyverse_4.sif

```
$ apptainer shell <apptainer_args> <container_url>:<container_version>
```

```
Apptainer> |
```



— enter —



```
$ apptainer run <args> <container_url>:<container_version> <program> <args>
```

```
$ apptainer run docker://rocker/tidyverse:4 R --version
```



— run —



```
R version 4.4.3 (2025-02-28) – "Trophy Case"
```



```
$ snakemake -c 1 --use-singularity --singularity-args <args>
```



```
rule rjob:
  container:
    "docker://rocker/tidyverse:4"
  shell:
    ""

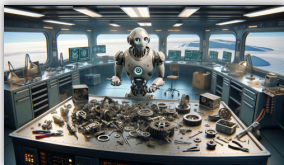
  Rscript R/script.R
  ""
```



```
rule rjob:
  container:
    "$HOME/tidyverse_4.sif"
  shell:
    ""

  Rscript R/script.R
  ""
```





scripting practices



version control



focus on projects



workflow managers



encapsulation of environments

DFG



DFG-SCHWERPUNKTPROGRAMM 1158

ANTARKTIS FORSCHUNG

mit vergleichenden Untersuchungen in arktischen Eisgebieten



ANTARCTIC

ANTARCTIC

ANTARCTIC

created using dalle

References

- Hench, K. *et al.* (2022). Rapid radiation in a highly diverse marine environment. *Proceedings of the National Academy of Sciences*, 119(4).